

FISH overhead campaigns — table collection

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List of Tables

1	Isaac ROS tracing overhead: throughput (fps, mean±std over 3 runs, measured by <code>ros2_benchmark</code> itself) per mode. Δ vs. the untraced baseline.	2
2	Autoware <code>logging_simulator</code> overhead: Autoware’s own internal measurements (<code>/diagnostics</code>), mean±std over 3 runs per mode; Δ vs. baseline. Zero topic-state-monitor violations in every mode.	2
3	Trace health across the overhead campaigns.	2
4	Isaac ROS, LTTng-only vs. untraced baseline over the peak-throughput test (3-run means, all metrics reported by <code>ros2_benchmark</code> itself), sorted by LTTng event rate. Cell: baseline value over $\Delta\%$	3
5	Isaac ROS, LTTng+nsys vs. untraced baseline over the peak-throughput test (3-run means, all metrics reported by <code>ros2_benchmark</code> itself), sorted by LTTng event rate. Cell: baseline value over $\Delta\%$	3
6	Event rate vs. throughput drop per benchmark.	4
7	Per-callback overhead, Isaac ROS (all benchmarks), LTTng-only (ref: best observed mean). Bars span min–max over 3 repetitions.	5
8	Per-callback overhead, Isaac ROS (all benchmarks), LTTng+nsys (ref: LTTng-only mean). Bars span min–max over 3 repetitions.	5
9	Per-callback overhead, Autoware, LTTng-only (ref: best observed mean). Bars span min–max over 3 repetitions.	6
10	Per-callback overhead, Autoware, LTTng+nsys (ref: LTTng-only mean). Bars span min–max over 3 repetitions.	6

1 Campaign summaries

Table 1: Isaac ROS tracing overhead: throughput (fps, mean $\pm$ std over 3 runs, measured by `ros2_benchmark` itself) per mode. Δ vs. the untraced baseline.

benchmark	baseline	LTtng-only	Δ	LTtng+nsys	Δ	jitter _{base} [ms]
apriltag	100.9 $\pm$ 0.1	100.4 $\pm$ 0.2	-0.5%	94.9 $\pm$ 0.2	-6.0%	0.2
bi3d	31.5 $\pm$ 0.3	31.8 $\pm$ 0.1	+1.1%	33.3 $\pm$ 3.0	+5.9%	11.4
bi3d_freespace	899.8 $\pm$ 0.0	899.8 $\pm$ 0.0	-0.0%	893.9 $\pm$ 7.4	-0.7%	1.0
dnn_image_encoder	9.1 $\pm$ 0.1	8.3 $\pm$ 0.2	-8.5%	8.2 $\pm$ 0.6	-9.5%	34.6
dope	7.6 $\pm$ 0.3	6.0 $\pm$ 0.3	-20.4%	5.9 $\pm$ 0.8	-22.0%	76.4
ess	45.8 $\pm$ 0.0	45.9 $\pm$ 0.0	+0.2%	45.7 $\pm$ 0.1	-0.2%	0.5
image_proc	423.0 $\pm$ 20.6	436.0 $\pm$ 14.7	+3.1%	411.5 $\pm$ 1.1	-2.7%	0.4
nvblox	7.3 $\pm$ 0.1	7.2 $\pm$ 0.1	-1.8%	7.1 $\pm$ 0.1	-2.6%	36.8
segformer	9.9 $\pm$ 1.2	7.3 $\pm$ 0.5	-26.3%	8.7 $\pm$ 0.2	-12.9%	30.3
stereo_image_proc	67.8 $\pm$ 0.4	69.1 $\pm$ 0.1	+1.9%	67.9 $\pm$ 0.3	+0.1%	0.4
tensor_rt	159.2 $\pm$ 0.3	158.1 $\pm$ 0.1	-0.7%	154.3 $\pm$ 0.2	-3.1%	1.0
unet	141.2 $\pm$ 0.1	116.8 $\pm$ 5.9	-17.3%	90.2 $\pm$ 6.3	-36.2%	1.6
visual_slam	33.6 $\pm$ 0.1	33.8 $\pm$ 0.1	+0.6%	33.9 $\pm$ 0.1	+0.7%	9.3

Table 2: Autoware `logging_simulator` overhead: Autoware’s own internal measurements (`/diagnostics`), mean $\pm$ std over 3 runs per mode; Δ vs. baseline. Zero topic-state-monitor violations in every mode.

metric	baseline	lttng-only (Δ)	lttng+nsys (Δ)
CPU cores (busiest 60 s)	3.86 $\pm$ 0.03	7.46 $\pm$ 0.10 (+93.4%)	6.63 $\pm$ 0.01 (+71.8%)
objects rate [Hz]	9.74 $\pm$ 0.11	9.78 $\pm$ 0.06 (+0.4%)	9.67 $\pm$ 0.06 (-0.7%)
obstacle pc rate [Hz]	9.83 $\pm$ 0.11	9.25 $\pm$ 0.48 (-5.9%)	9.53 $\pm$ 0.10 (-3.1%)
concat processing [ms]	6.44 $\pm$ 0.15	11.20 $\pm$ 0.40 (+73.9%)	11.35 $\pm$ 0.31 (+76.2%)
concat pipeline lat [ms]	182.70 $\pm$ 0.12	196.08 $\pm$ 0.15 (+7.3%)	197.45 $\pm$ 1.15 (+8.1%)
centerpoint proc [ms] (GPU)	22.69 $\pm$ 0.13	22.82 $\pm$ 0.71 (+0.6%)	23.03 $\pm$ 0.11 (+1.5%)
Total Latency [ms]	380.77 $\pm$ 6.05	410.25 $\pm$ 11.04 (+7.7%)	410.69 $\pm$ 10.59 (+7.9%)
multi_object_tracker [ms]	135.96 $\pm$ 6.62	159.99 $\pm$ 7.88 (+17.7%)	160.39 $\pm$ 6.97 (+18.0%)
tsm NotReceived (startup)	19.00 $\pm$ 8.00	35.33 $\pm$ 15.28 (+86.0%)	23.00 $\pm$ 9.54 (+21.1%)
tsm violations (post-init)	0$\pm$0	0$\pm$0	0$\pm$0

Table 3: Trace health across the overhead campaigns.

campaign	traced runs	events	discarded
Isaac ROS (13 benchmarks)	90	60.1 M	3 178 (0.005%)
Autoware logging_simulator	6	30.5 M	0

2 Full ros2_benchmark metric matrices

Table 4: Isaac ROS, LTTng-only vs. untraced baseline over the peak-throughput test (3-run means, all metrics reported by `ros2_benchmark` itself), sorted by LTTng event rate. Cell: baseline value over $\Delta\%$.

benchmark	events [kev/s]	fps [fps]	play fps [fps]	sent [frame]	missed [frame]	recv dur [ms]	first e2e [ms]	last e2e [ms]	jit mean [ms]	jit sd [ms]	jit max [ms]	cpu mean [%]	cpu max [%]
nvblox	0.2	7.30 -1.8%	135.1 +2.3%	818.0 +0.2%	749.3 +0.5%	9767 -0.6%	1220 +2.5%	4897 +1.7%	36.8 -7.2%	29.2 -4.9%	110.6 -12.3%	8.57 +42.1%	41.1 +1.3%
dope	0.5	7.17 -16.0%	10.2 +0.0%	50.0 +0.0%	15.7 +46.1%	4789 -5.3%	98.4 +90.2%	-1.77 +6310.1%	100.5 +40.8%	104.3 +30.5%	494.4 +28.1%	1.56 +43.2%	15.8 +84.2%
ess	0.6	82.6 +0.3%	84.6 +0.0%	421.0 +0.0%	6.56 -23.7%	5018 +0.0%	13.8 -1.4%	53.4 +3.7%	0.44 +3.4%	1.04 +20.7%	24.1 +0.1%	4.95 -40.9%	25.0 +17.8%
apriltag	0.7	303.2 -0.7%	305.2 +0.0%	1525 +0.0%	10.1 +107.7%	4997 -0.0%	3.90 -0.8%	4.60 -9.5%	0.09 +40.1%	0.12 +90.3%	2.25 +54.8%	2.25 +11.2%	17.5 +90.5%
tensor_rt	1.4	536.2 -1.4%	543.6 -1.5%	2717 -62.4%	26.9 -62.4%	5017 -0.1%	4.43 -3.8%	23.5 -14.0%	0.87 -8.9%	0.45 +3.1%	3.77 +9.5%	3.82 +6.5%	25.8 +112.9%
stereo_image_proc	1.6	170.5 +3.0%	174.3 +3.5%	870.3 +3.4%	17.1 +23.4%	5004 -0.0%	6.96 -2.3%	17.4 +9.2%	0.23 +4.1%	0.36 +25.2%	8.05 +40.3%	3.02 +9.3%	20.0 +68.1%
unet	5.7	464.3 -21.0%	475.4 -5.4%	2375 +776.0%	45.3 +776.0%	5019 +0.2%	8.67 +6.2%	31.4 +28.0%	0.62 +69.5%	0.56 +69.2%	5.01 +125.6%	8.86 +24.8%	49.4 +28.9%
dnn_image_encoder	6.4	9.39 -11.4%	10.2 +0.0%	50.0 +0.0%	4.56 +126.8%	4857 -1.6%	41.5 +133.5%	8.76 -388.7%	26.4 +131.1%	48.8 +62.4%	231.3 +58.3%	0.69 +55.2%	15.0 +194.4%
bi3d_freespace	7.4	3499 -0.0%	3498 -0.0%	17490 +0.0%	0.00 $\pm 0.0\%$	4999 +0.0%	0.92 -0.1%	0.58 +2.6%	0.15 -1.5%	0.12 -1.6%	1.46 -32.6%	3.05 +13.5%	20.0 +33.3%
image_proc	7.7	1592 +3.3%	1600 +3.4%	7997 +3.4%	41.1 +28.9%	4999 +0.0%	1.63 -1.9%	1.07 +2.8%	0.15 -2.2%	0.13 +1.4%	1.72 +32.5%	5.00 +21.6%	29.2 +92.4%

Table 5: Isaac ROS, LTTng+nsys vs. untraced baseline over the peak-throughput test (3-run means, all metrics reported by `ros2_benchmark` itself), sorted by LTTng event rate. Cell: baseline value over $\Delta\%$.

benchmark	events [kev/s]	fps [fps]	play fps [fps]	sent [frame]	missed [frame]	recv dur [ms]	first e2e [ms]	last e2e [ms]	jit mean [ms]	jit sd [ms]	jit max [ms]	cpu mean [%]	cpu max [%]
nvblox	0.2	7.30 -2.6%	135.1 +6.9%	818.0 -0.1%	749.3 +0.4%	9767 -3.5%	1220 +0.8%	4897 +1.9%	36.8 -6.7%	29.2 -2.0%	110.6 -15.0%	8.57 +5.4%	41.1 -10.1%
dope	0.5	7.17 -17.6%	10.2 +0.0%	50.0 +0.0%	15.7 +43.3%	4789 -3.7%	98.4 +57.2%	-1.77 +4992.7%	100.5 +39.2%	104.3 +31.7%	494.4 +1.4%	1.56 +19.5%	15.8 +91.2%
ess	0.5	82.6 -0.5%	84.6 +0.0%	421.0 +0.0%	6.56 +30.5%	5018 +0.1%	13.8 +1.9%	53.4 +7.9%	0.44 +15.3%	1.04 +13.1%	24.1 +0.5%	4.95 -10.5%	25.0 +90.0%
apriltag	0.6	303.2 -8.0%	305.2 -6.0%	1525 -6.1%	10.1 +284.6%	4997 -0.0%	3.90 +7.7%	4.60 +42.3%	0.09 +47.2%	0.12 +38.5%	2.25 +30.9%	2.25 +11.6%	17.5 +66.7%
tensor_rt	1.2	536.2 -3.7%	543.6 -3.6%	2717 -3.6%	26.9 +10.3%	5017 -0.1%	4.43 -0.7%	23.5 -17.7%	0.87 +0.2%	0.45 +64.1%	3.77 +292.9%	3.82 -9.2%	25.8 +238.7%
stereo_image_proc	1.5	170.5 +0.2%	174.3 +2.0%	870.3 +2.0%	17.1 +90.9%	5004 -0.1%	6.96 +1.7%	17.4 -1.4%	0.23 +3.9%	0.36 -24.3%	8.05 -2.7%	3.02 +14.1%	20.0 +108.3%
unet	3.3	464.3 -44.0%	475.4 -42.3%	2375 -42.4%	45.3 +48.5%	5019 -0.3%	8.67 +14.1%	31.4 -31.5%	0.62 +194.3%	0.56 +199.4%	5.01 +261.4%	8.86 -4.1%	49.4 +77.5%
dnn_image_encoder	4.9	9.39 -9.6%	10.2 +0.0%	50.0 +0.0%	4.56 +109.8%	4857 -2.1%	41.5 +160.4%	8.76 -508.5%	26.4 +127.9%	48.8 +61.8%	231.3 +44.0%	0.69 +47.0%	15.0 +174.1%
image_proc	5.8	1592 -2.9%	1600 -1.7%	7997 -1.7%	41.1 +223.0%	4999 +0.0%	1.63 +5.4%	1.07 +64.0%	0.15 +18.6%	0.13 +25.5%	1.72 +98.6%	5.00 +46.5%	29.2 +214.3%
bi3d_freespace	6.1	3499 -0.7%	3498 -0.5%	17490 -0.5%	0.00 →22.9	4999 +0.0%	0.92 +27.6%	0.58 +150.0%	0.15 -9.5%	0.12 +4.2%	1.46 +151.1%	3.05 +32.2%	20.0 +279.2%

3 Overhead vs. event rate

Table 6: Event rate vs. throughput drop per benchmark.

benchmark	LTTng-only		LTTng+nsys	
	[kev/s]	drop [%]	[kev/s]	drop [%]
apriltag	0.7	0.5	0.6	6.0
bi3d_freespace	7.4	0.0	6.1	0.7
dnn_image_encoder	6.4	8.5	4.9	9.5
dope	0.5	20.4	0.5	22.0
ess	0.6	-0.2	0.5	0.2
image_proc	7.7	-3.1	5.8	2.7
nvblox	0.2	1.8	0.2	2.6
stereo_image_proc	1.6	-1.9	1.5	-0.1
tensor_rt	1.4	0.7	1.2	3.1
unet	5.7	17.3	3.3	36.2

4 Per-callback overhead (top/middle/bottom 10 by max)

Table 7: Per-callback overhead, Isaac ROS (all benchmarks), LTTng-only (ref: best observed mean). Bars span min–max over 3 repetitions.

callback (entity (node))	GPU	min [μ s]	mean [μ s]	max [μ s]
[nvblox] sub:/r2b/hawk_0_right_rgb_camera_info (VisualSlamNode)	✓	0.0	1069.4	2890.9
[nvblox] sub:/r2b/hawk_0_right_mono_image (VisualSlamNode)	✓	0.0	1184.9	1854.2
[dope] sub:/r2b/image (resize_node)	✓	614.4	1066.1	1692.9
[nvblox] sub:/r2b/hawk_0_left_mono_image (VisualSlamNode)	✓	0.0	535.5	1571.7
[dope] sub:/r2b/crop/image (image_to_tensor)	✓	548.1	900.6	1197.5
[dope] sub:/r2b/image_tensor (normalize_node)	✓	245.6	361.1	469.2
[nvblox] sub:/r2b/hawk_0_left_rgb_image (PrepLeftImageFormatConverterNode)	✓	0.0	62.9	180.9
[nvblox] sub:/r2b/depth (NvbloxNode)	✓	0.0	50.2	142.8
[nvblox] sub:/r2b/hawk_0_right_rgb_image (PrepRightImageFormatConverterNode)	✓	0.0	50.2	139.7
[nvblox] sub:/r2b/left/rgb (NvbloxNode)	✓	0.0	53.0	132.9
[bi3d_freespace] tmr:1000000000ns (PrepRightResizeNode)	✓	0.0	1.5	2.4
[stereo_image_proc] waitable (PrepRightResizeNode)		0.0	0.9	2.4
[unet] sub:/parameter_events (NormalizeNode)		1.2	1.6	2.4
[dope] sub:/r2b/camera_info (resize_node)	✓	0.0	1.4	2.4
[dope] sub:/parameter_events (normalize_node)		0.8	1.4	2.4
[dope] waitable (interleaved_to_planar_node)		0.0	1.2	2.3
[dnn_image_encoder] sub:/parameter_events (normalize_node)		0.5	1.2	2.3
[bi3d_freespace] sub:/parameter_events (PrepRightResizeNode)		1.5	1.9	2.2
[nvblox] sub:/parameter_events (PrepRightImageFormatConverterNode)		0.5	1.2	2.2
[nvblox] sub:/parameter_events (PlaybackNode)		0.0	1.0	2.1
[dnn_image_encoder] sub:/r2b/resize/image (image_format_converter_node)		0.0	0.1	0.2
[dope] waitable (reshape_node)		0.0	0.1	0.2
[stereo_image_proc] sub:/r2b/disparity (MonitorNode)		0.0	0.1	0.2
[bi3d_freespace] sub:/r2b/freespace_segmentation/occup... (MonitorNode)	✓	0.0	0.1	0.2
[dnn_image_encoder] sub:/r2b/camera_info (resize_node)		0.0	0.1	0.1
[dnn_image_encoder] tmr:100000000ns (image_format_converter_node)		0.0	0.1	0.1
[dnn_image_encoder] sub:/r2b/output (MonitorNode)	✓	0.0	0.1	0.1
[stereo_image_proc] tmr:100000000ns (PrepLeftResizeNode)		0.0	0.1	0.1
[dnn_image_encoder] sub:/r2b/image (resize_node)		0.0	0.0	0.0
[dnn_image_encoder] sub:/r2b/normalized_tensor (interleaved_to_planar_node)		0.0	0.0	0.0

Table 8: Per-callback overhead, Isaac ROS (all benchmarks), LTTng+nsys (ref: LTTng-only mean). Bars span min–max over 3 repetitions.

callback (entity (node))	GPU	min [μ s]	mean [μ s]	max [μ s]
[apriltag] tmr:1000000000ns (PrepResizeNode)		1872.5	1934.0	1978.5
[stereo_image_proc] tmr:1000000000ns (PrepLeftResizeNode)	✓	1839.3	1861.1	1880.1
[stereo_image_proc] tmr:1000000000ns (PrepRightResizeNode)	✓	1744.3	1793.3	1827.4
[dope] tmr:1000000000ns (resize_node)	✓	1683.8	1743.7	1795.2
[dnn_image_encoder] tmr:1000000000ns (resize_node)	✓	1696.9	1730.0	1780.2
[image_proc] tmr:1000000000ns (RectifyNode)	✓	1691.6	1740.9	1777.7
[bi3d_freespace] tmr:1000000000ns (PrepLeftResizeNode)	✓	1502.6	1561.7	1625.3
[bi3d_freespace] tmr:1000000000ns (PrepRightResizeNode)	✓	1493.0	1530.9	1596.7
[dope] tmr:1000000000ns (image_format_converter_node)	✓	1414.8	1465.1	1493.4
[dnn_image_encoder] tmr:1000000000ns (image_format_converter_node)	✓	1356.5	1397.0	1450.3
[stereo_image_proc] tmr:1000000000ns (PlaybackNode)		1.9	2.5	2.8
[dnn_image_encoder] sub:/r2b/output (MonitorNode)	✓	1.9	2.2	2.7
[dnn_image_encoder] sub:/r2b/image (resize_node)		2.4	2.6	2.7
[image_proc] sub:/parameter_events (RectifyNode)		-0.5	0.7	2.7
[image_proc] sub:/r2b/image (RectifyNode)		-0.3	1.6	2.7
[dope] tmr:1000000000ns (resize_node)		0.8	1.7	2.6
[dope] sub:/r2b/tensor_pub (TensorRTNode)	✓	2.1	2.3	2.4
[unet] sub:/r2b/tensor (NormalizeNode)	✓	-39.6	-19.7	2.4
[dope] tmr:1000000000ns (DopeDecoder)		1.4	2.0	2.3
[tensor_rt] tmr:1000000000ns (PlaybackNode)		-3.4	0.3	2.3
[stereo_image_proc] sub:/parameter_events (PrepRightResizeNode)		-2.5	-2.0	-1.3
[unet] sub:/parameter_events (ImageFormatConverter)		-1.9	-1.6	-1.4
[nvblox] sub:/parameter_events (MonitorNode1)		-2.2	-1.9	-1.5
[unet] sub:/parameter_events (TensorRTNode)		-3.6	-2.5	-1.6
[dnn_image_encoder] sub:/parameter_events (interleaved_to_planar_node)		-3.4	-2.9	-2.1
[tensor_rt] tmr:1000000000ns (TensorRTNode)		-10.5	-8.5	-6.0
[nvblox] sub:/r2b/hawk_0_left_rgb_image (PrepLeftImageFormatConverterNode)	✓	-45.3	-21.6	-6.1
[unet] waitable (ResizeNode)	✓	-30.3	-29.5	-28.1
[nvblox] sub:/r2b/hawk_0_left_mono_image (VisualSlamNode)	✓	-518.7	-514.5	-506.6
[dope] sub:/r2b/image (resize_node)	✓	-1066.1	-870.4	-634.7

Table 9: Per-callback overhead, Autoware, LTTng-only (ref: best observed mean). Bars span min–max over 3 repetitions.

callback (entity (node))	GPU	min [μ s]	mean [μ s]	max [μ s]
tmr:100000000ns (?)		0.0	1244.1	3420.9
sub:/perception/obstacle_segmentation... (voxel_based_compare_map_filter)		0.0	965.3	1710.8
tmr:100000000ns (collision_detector)		0.0	807.2	1661.8
sub:/perception/object_recognition/ob... (perception_analytics_publisher)		0.0	422.5	860.2
tmr:100000000ns (voxel_based_compare_map_filter)		229.1	494.3	748.6
tmr:100000000ns (autonomous_emergency_braking)		8.7	254.3	719.8
sub:/sensing/lidar/top/velodyne_packe... (velodyne_ros_wrapper_node)		0.0	307.4	596.2
tmr:20000000ns (ekf_localizer)		0.0	169.0	445.8
sub:/sensing/lidar/top/pointcloud_bef... (concatenate_data)		0.0	175.9	362.6
sub:/localization/util/downsample/poi... (ndt_scan_matcher)		0.0	164.7	342.6
sub:/sensing/imu/tamagawa/imu_raw (gyro_bias_scale_validator)		0.0	5.3	8.2
sub:/parameter_events (autoware_initial_pose_adaptor)		0.0	4.9	8.2
sub:/clock (scenario_selector)		5.0	6.1	8.2
sub:/clock (multi_object_tracker)		0.0	3.0	8.1
sub:/clock (velocity_smoother)		1.6	4.9	8.1
sub:/parameter_events (traffic_light_fine_detector)		0.1	2.8	8.0
sub:/parameter_events (lane_departure_checker_node)		0.0	3.8	8.0
tmr:100000000ns (processing_time_checker)		1.2	5.0	7.9
sub:/localization/twist_estimator/twi... (ekf_localizer)		0.6	4.8	7.9
sub:/parameter_events (crop_box_filter_mirror)		0.9	3.8	7.9
sub:/perception/object_recognition/pr... (processing_time_checker)		0.2	0.2	0.3
tmr:200000000ns (autoware_routing_adaptor)		0.0	0.1	0.3
sub:/system/operation_mode/availabili... (mrm_handler)		0.0	0.1	0.3
sub:/diagnostics (component)		0.0	0.1	0.3
sub:/perception/object_recognition/tr... (perception_analytics_publisher)		0.0	0.1	0.2
sub:/localization/kinematic_state (manual_lane_change_handler)		0.0	0.1	0.2
sub:/control/external_cmd_selector/cu... (autoware_operation_mode_transition_manager)		0.1	0.1	0.2
sub:/perception/object_recognition/pr... (perception_analytics_publisher)		0.1	0.1	0.1
sub:/localization/acceleration (vehicle_cmd_gate)		0.0	0.0	0.1
sub:/vehicle/status/steering_status (vehicle_cmd_gate)		0.0	0.0	0.1

Table 10: Per-callback overhead, Autoware, LTTng+nsys (ref: LTTng-only mean). Bars span min–max over 3 repetitions.

callback (entity (node))	GPU	min [μ s]	mean [μ s]	max [μ s]
sub:/perception/obstacle_segmentation... (voxel_based_compare_map_filter)		492.9	1845.9	3793.9
sub:/perception/object_recognition/ob... (perception_analytics_publisher)		33.8	900.9	1658.8
tmr:100000000ns (autonomous_emergency_braking)		-254.3	217.1	1137.6
sub:/perception/object_recognition/de... (multi_object_tracker)		1086.0	1092.9	1101.3
tmr:100000000ns (collision_detector)		-155.3	242.6	741.2
tmr:100000000ns (voxel_based_compare_map_filter)		-494.3	230.6	626.7
sub:/sensing/lidar/top/pointcloud_bef... (concatenate_data)		265.2	339.5	482.2
sub:/localization/kinematic_state (voxel_based_compare_map_filter)		-11.0	209.6	420.1
sub:/localization/util/downsample/poi... (ndt_scan_matcher)		136.4	305.3	410.7
sub:/perception/object_recognition/tr... (map_based_prediction)		-81.7	130.8	326.8
sub:/clock (converter)		-0.8	2.0	4.8
sub:/parameter_events (planning)		-7.0	-1.2	4.8
sub:/clock (ekf_localizer)		1.1	3.0	4.8
sub:/clock (traffic_light_image_decompressor)		0.4	2.4	4.8
sub:/clock (vector_map_tf_generator)		0.3	2.7	4.7
sub:/clock (map_hash_generator)		-6.3	0.0	4.7
sub:/clock (mrm_emergency_stop_operator)		-1.6	1.7	4.5
tmr:200000000ns (planning)		-12.8	-7.0	4.4
sub:/parameter_events (traffic_light_map_visualizer)		-12.6	-5.8	4.4
sub:/clock (glog_component)		-1.3	1.0	4.3
sub:/parameter_events (motion_velocity_planner)		-16.9	-13.7	-12.0
sub:/parameter_events (perception)		-17.8	-15.7	-12.4
tmr:100000000ns (traffic_light_camera_info_relay)		-31.1	-25.3	-14.7
tmr:100000000ns (vehicle_status)		-27.8	-20.0	-15.4
sub:/parameter_events (vector_map_tf_generator)		-25.0	-20.3	-16.9
tmr:100000000ns (autoware_state)		-54.0	-40.7	-18.8
tmr:100000000ns (traffic_light_camera_info_relay)		-35.3	-30.0	-27.3
sub:/sensing/lidar/left/velodyne_pack... (velodyne_ros_wrapper_node)		-42.8	-34.5	-27.6
tmr:100000000ns (trajectory_relay)		-47.1	-39.8	-28.2
tmr:100000000ns (?)		-991.4	-495.6	-241.5